

This technical reference summarizes the analytical mapping between Monin–Obukhov Similarity Theory (MOST) flux-profile functions ϕ_m, ϕ_h and gradient Richardson number stability functions $S_m(Ri), S_h(Ri)$ across stable, unstable, and surface-discretized regimes.

Fundamental Coupling Relations

- **Momentum Stability Function:** $S_m(Ri) = \phi_m^{-2}(\zeta)$
- **Heat/Scalar Stability Function:** $S_h(Ri) = [\phi_m(\zeta)\phi_h(\zeta)]^{-1} = \frac{S_m^{1/2}(Ri)}{\phi_h(\zeta)}$
- **Exact Inversion Mapping ($\zeta \leftrightarrow Ri$):**

$$\zeta = Ri \frac{S_h(Ri)}{S_m^{3/2}(Ri)}$$

(Corrects the S_m exponent from 1 to 3/2 for internal consistency when $Pr_t \neq 1$).

Analytical Solutions Across Stratification Regimes

Regime / Parameterization	Governing Profile Exponents	Closed-Form Stability Functions $S_m(Ri), S_h(Ri)$
Stable Linear Heat	$\phi_m = 1 + \beta_m \zeta,$ $\phi_h = 1 + \beta_h \zeta$	$S_m(Ri) = \left[\frac{(1+Ri\beta_h) - \sqrt{(1+Ri\beta_h)^2 - 4Ri(\beta_h - \beta_m)}}{2Ri(\beta_h - \beta_m)} \right]^2$ $S_h(Ri) = \frac{x^2(1-x)}{\beta_m Ri}$ where $x \equiv S_m^{1/2}$
Stable Equal Parameters	$\beta_m = \beta_h = \beta$ ($Pr_t = 1, Ri_c = 1/\beta$)	$S_m(Ri) = S_h(Ri) = (1 - \beta Ri)^2 = \left(1 - \frac{Ri}{Ri_c}\right)^2$
Unstable Businger–Dyer	$\alpha_m = 1/4, \alpha_h = 1/2$ Master Variable $x \equiv S_m^2$	Master Cubic: $x^3 - 2x^2 + (1 - Ri^2\beta_m\beta_h)x - Ri^2\beta_m(\beta_m - \beta_h) = 0$
Unstable Equal Coefficients	$\beta_m = \beta_h = 16$	$S_m(Ri) = (1 - 16Ri)^{1/2}$ $S_h(Ri) = (1 - 16Ri)^{3/4}$
		$S_m(Ri) = 1 - 2\beta_m Ri - \beta_m(\beta_m - 2\beta_h) Ri^2 +$

Near-Neutral Series	Expansion around $Ri \rightarrow 0$	$\mathcal{O}(Ri^3) S_h(Ri) = 1 - (\beta_m + \beta_h) Ri - (\beta_m^2 - 2\beta_h^2 + \gamma_m + \gamma_h) Ri^2 + \mathcal{O}(Ri^3)$
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Discrete Surface Layer & Microscale Thermal Roughness

Evaluating surface exchange in numerical models without non-linear iterations incorporates excess thermal resistance ($kB^{-1} \equiv \ln(z_{0m}/z_{0h})$) directly into a discretized layer Richardson number $Ri_{1/2}$:

- **Effective Height Scale (h_1):**

$$h_1 = \sqrt{z_1 z_{0m}} e^{\frac{1}{2}kB^{-1}} \frac{\left[\ln \left(\frac{z_1}{z_{0m}} \right) \right]^2}{\ln \left(\frac{z_1}{z_{0m}} \right) + kB^{-1}}$$

- **Mid-Layer Stability Parameter:** $Ri_{1/2} = \frac{g}{\theta_0} h_1 \frac{\theta_1 - \theta_g}{|\mathbf{V}_1|^2}$
- **Direct Non-Iterative Flux Equations:**

$$u_*^2 = \frac{\kappa^2 S_m(Ri_{1/2}) |\mathbf{V}_1|^2}{\left[\ln \left(\frac{z_1}{z_{0m}} \right) \right]^2}$$

$$u_* \theta_* = \frac{\kappa^2 S_h(Ri_{1/2}) |\mathbf{V}_1| (\theta_1 - \theta_g)}{\ln \left(\frac{z_1}{z_{0m}} \right) \left[\ln \left(\frac{z_1}{z_{0m}} \right) + kB^{-1} \right]}$$